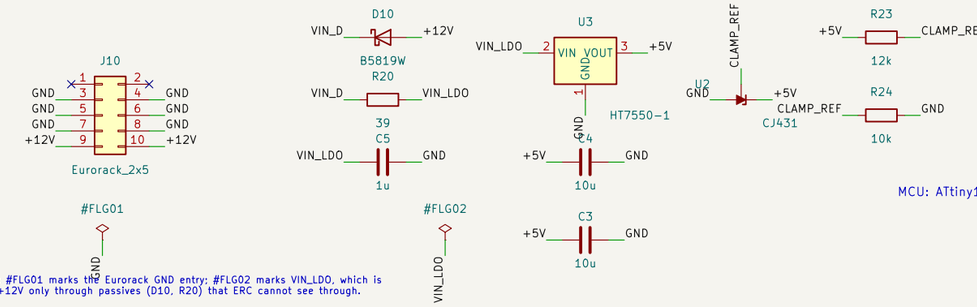
BIG MODULAR MIDI V2 — S3 schematic (5 V logic, mis-patch protected in/out). Circuit intent: docs/research.md 7

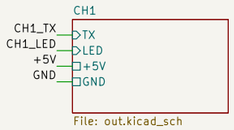
POWER: Eurorack +12V -> B5819W -> 39R -> HT7550-1 -> +5V. -12V (J10 pins 1-2) unused.

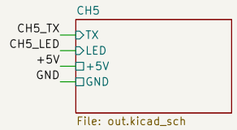
U2 +5V CLAMP: sinks output-clamp current above ~5.5 V (the LDO cannot sink).

OUTPUTS 1-8: sheets CH1-CH8 (out.kicad_sch): GPIO -> 1k -> TS tip (sleeve GND), BAT54SW clamp on the GPIO side; LED: GPIO -> 10k -> NPN, +5V -> LED -> 1k -> collector (Bread Modular gate_out)

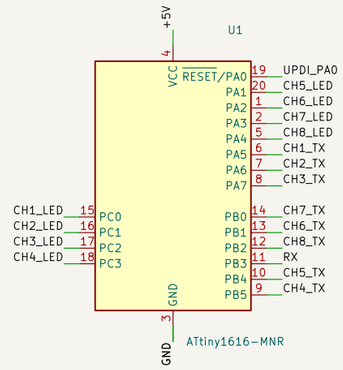


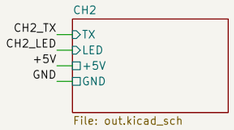
PWR_FLAG: #FLG01 marks the Eurorack GND entry; #FLG02 marks VIN_LDO, which is fed from +12V only through passives (D10, R20) that ERC cannot see through.

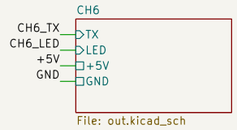




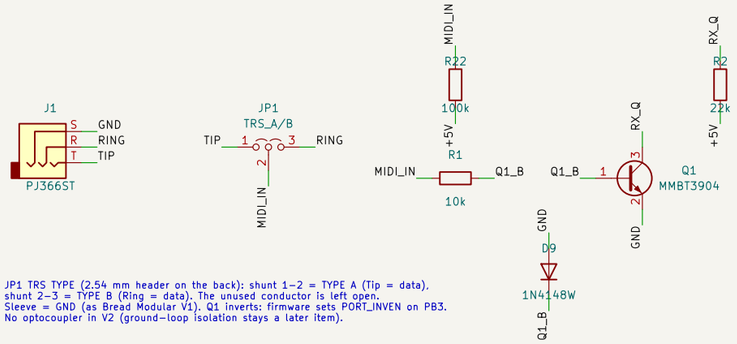
MCU: ATtiny1616 at 5 V, 20 MHz; pin map identical to Bread Modular

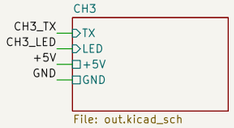


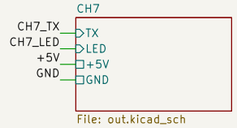




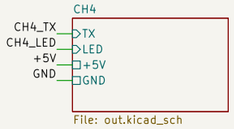
MIDI IN: NPN front end (3.3 V and 5 V senders, +/-12 V tolerant), no opto

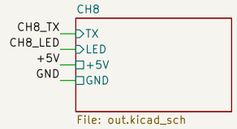






JP1 TRS TYPE (2.54 mm header on the back): shunt 1-2 = TYPE A (Tip = data), shunt 2-3 = TYPE B (Ring = data). The unused conductor is left open. Sleeve = GND (as Bread Modular V1). Q1 inverts: firmware sets PORT_INVEN on PB3. No optocoupler in V2 (ground-loop isolation stays a later item).





JP2 RX SOURCE (solder jumper on the back, shipped OPEN: bridge one pair by hand): Solder 1-2 = PROTECTED (Q1 drives PB3: firmware PORT_INVEN on PB3). Solder 2-3 = DIRECT debug path (MIDI_IN to PB3 as V1, unprotected: PORT_INVEN off). Exactly one bridge: never both, and open = PB3 floats.

UPDI J11: pin 1 UPDI, pin 2 GND (2.54 mm). Power the module from Eurorack while programming.

Circuit intent: docs/research.md; placement: docs/v2/geometry.json
V2 SCHEMATIC S4 — 5 V logic, mis-patch protected input and outputs, JP2 input bypass (open), LED

Big Modular

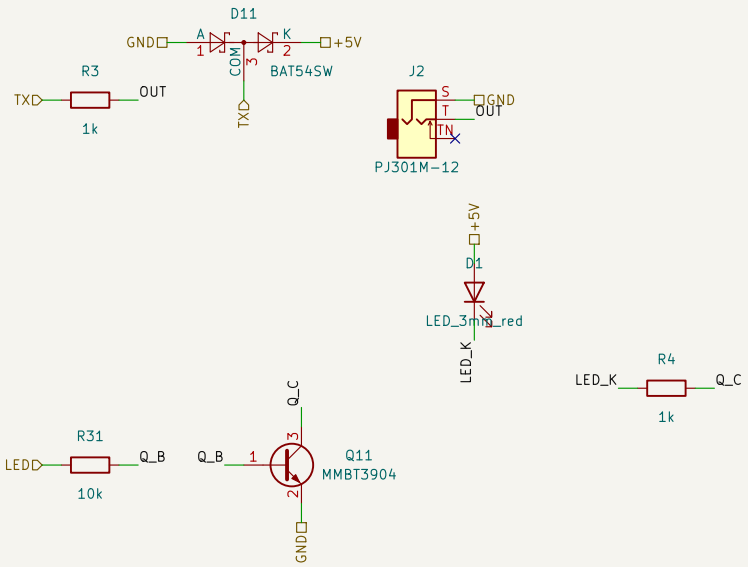
Sheet: /
File: midi.kicad_sch

Title: Big Modular — MIDI V2

Size: A3	Date:	**Rev: V2-S4**
KiCad E.D.A. 10.0.6		Id: 1/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

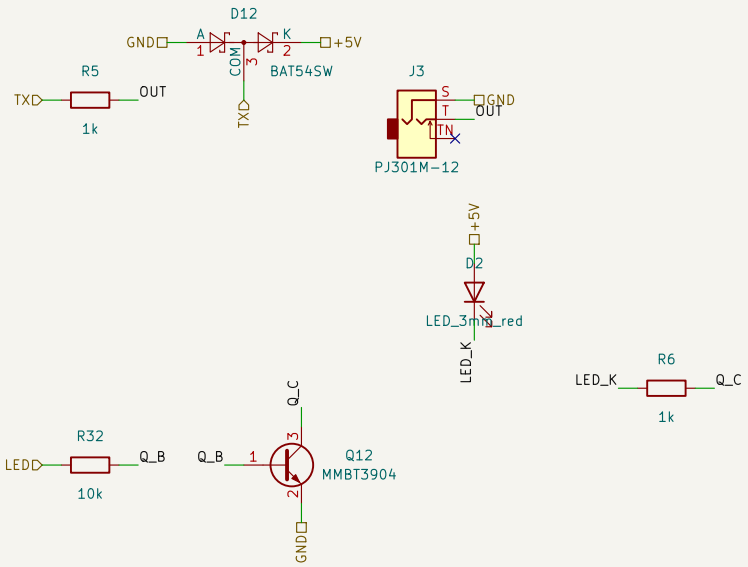
Sheet: /CH1/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 2/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

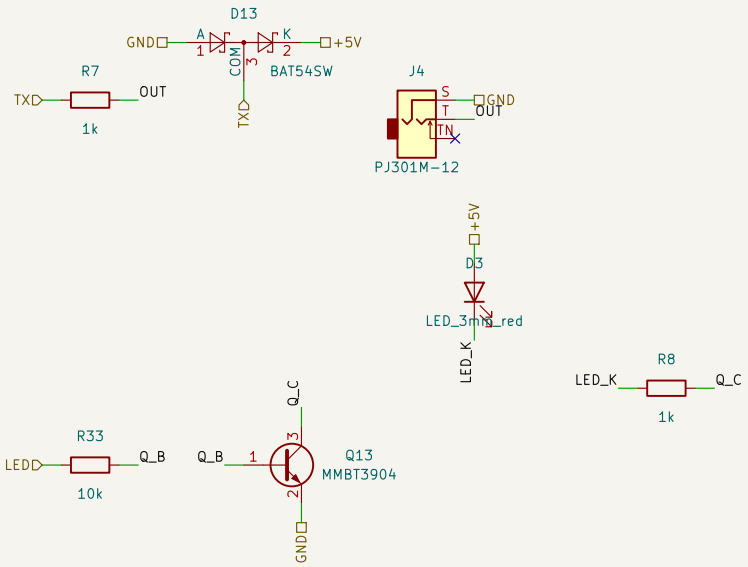
Sheet: /CH2/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 3/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

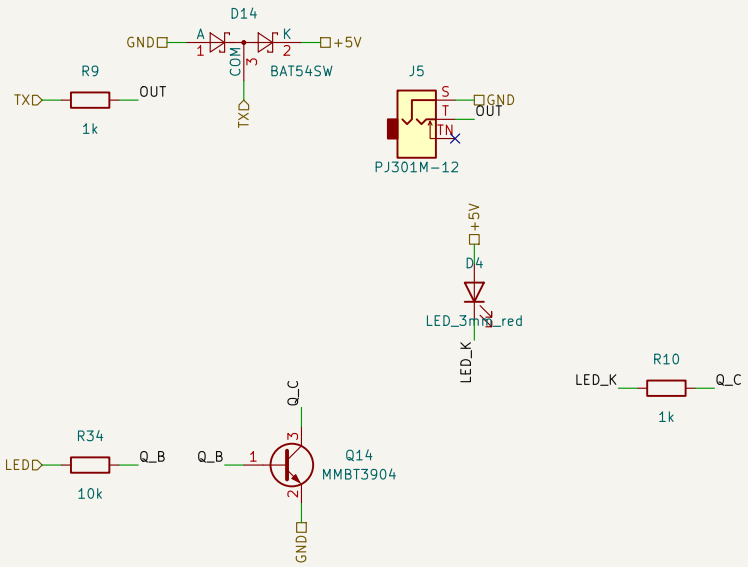
Sheet: /CH3/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 4/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

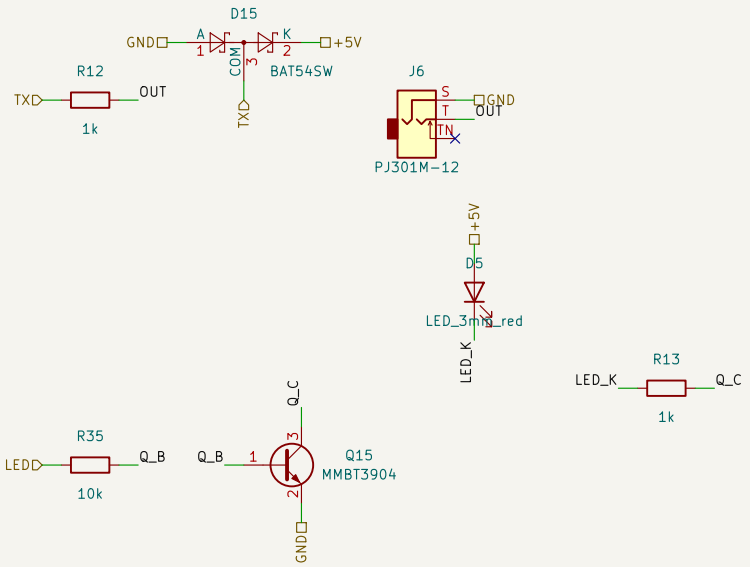
Sheet: /CH4/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 5/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

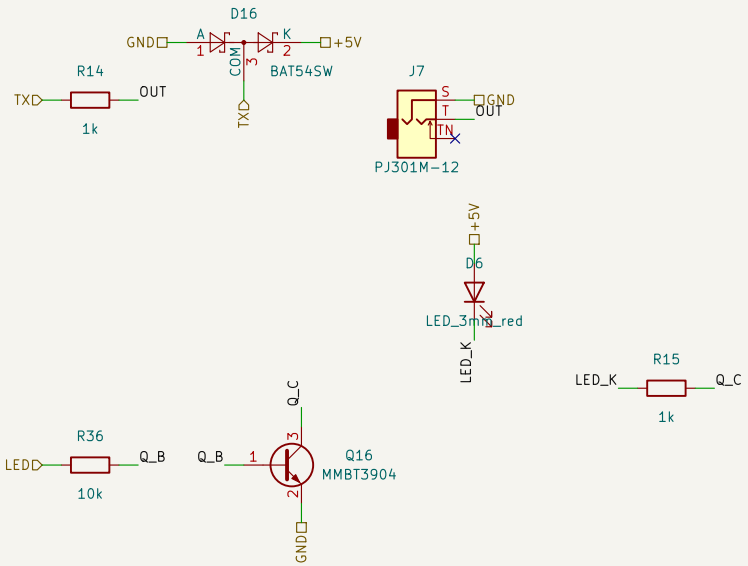
Sheet: /CH5/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 6/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

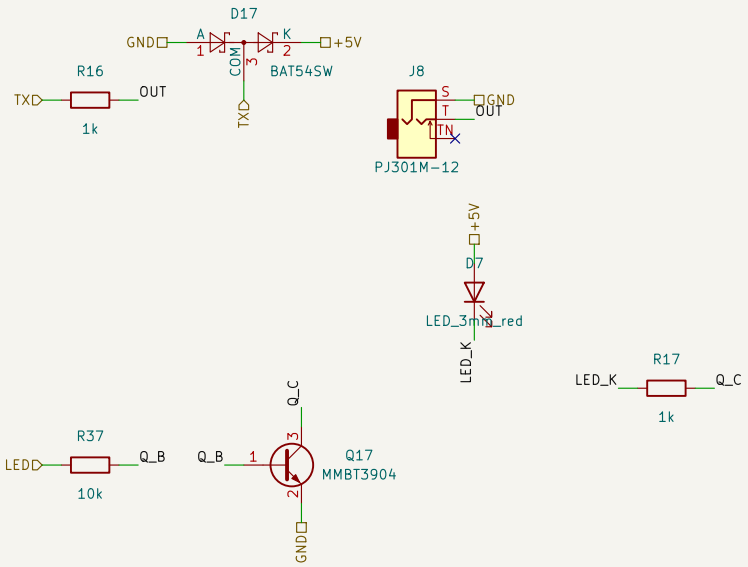
Sheet: /CH6/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 7/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

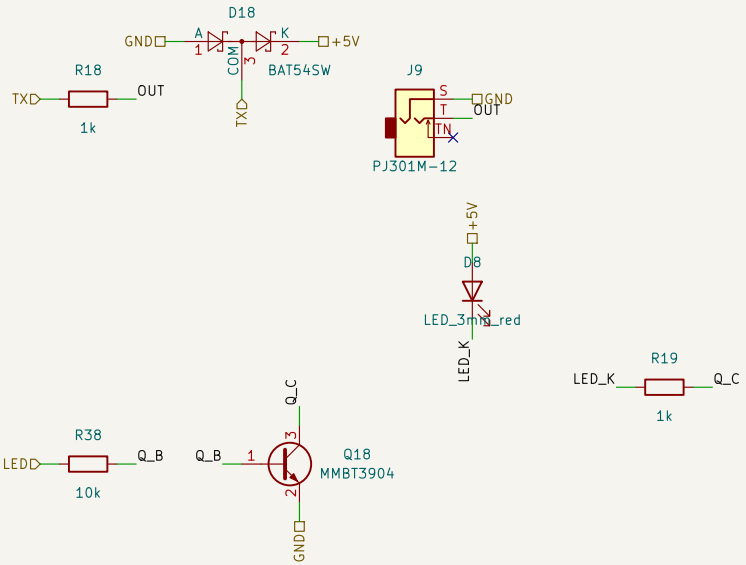
Sheet: /CH7/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 8/9

OUTPUT CHANNEL (instanced as CH1–CH8). TX = U1 output pin, LED = U1 LED pin.

MIDI out: TX –> 1k (0805) –> TS tip, sleeve GND; BAT54SW clamps TX to +5V/GND (research 7.3).



LED driver as Bread Modular gate_out (SS8050 there): the GPIO only drives the base through 10k (~0.4 mA); the ~3 mA LED current comes from +5V through the LED and 1k.

Circuit intent: docs/research.md 3.4, 7.3
Instanced as CH1–CH8 in midi.kicad_sch
Big Modular

Sheet: /CH8/
File: out.kicad_sch

Title: Big Modular – MIDI V2 output channel

Size: A4	Date:	**Rev: V2–S4**
KiCad E.D.A. 10.0.6		Id: 9/9